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MILITARY SPECIFICATION

REGULATOR, OXYGEN, AUTOMATIC PRESSURE BREATHING HIGH ALTITUDE, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope.- This specification covers automatic pressure breathing, high altitude oxygen regulators for installation in survival kits.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

BB-O-925 Oxygen, 99.5 Percent, Gas and Liquid

Military

MIL-P-116 Preservation, Methods Of
MIL-D-5028 Drawings and Data Lists; Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-C-5541 Chemical Films for Aluminum and Aluminum Alloys
MIL-N-6011 Nitrogen; Liquid and Gas
MIL-P-7105 Pipe Threads; Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742 Screw Threads, Standard, Aeronautical
MIL-P-7936 Parts and Equipment, Aeronautical, Preparation For Delivery
MIL-A-8625 Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-9107 Test Reports; Preparation Of

FSC 6680

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STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MS33586	Dissimilar Metals, Definition Of

PUBLICATIONS

Air Force/Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provision for preproduction testing.

3.2 Selection of Specifications and Standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to ANA Bulletin 143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, etc., having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard Parts.- With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

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3.3 Materials.- Any material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used. The materials shall be entirely suitable for the purpose intended.

3.3.1 Nonferrous Materials.- Nonferrous materials shall be used for all parts of the regulator except where ferrous materials are essential.

3.3.2 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.3 Elastomers.- All elastomers of natural rubber or any synthetic rubber other than certain silicone types (which are to be approved by the procuring activity) shall have been manufactured not more than 12 months prior to the date of regulator manufacture. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. The diaphragm shall be fabricated of an approved silicone compound or silicone coated cloth. All elastomers shall be free from foreign agents which may cause objectionalbe or intolerable odors.

3.3.4 Protective Treatment.- When materials are used in the construction of the regulator that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design.- The regulator shall be of the automatic type providing both automatic pressure breathing and altitude suit pressure. The regulator shall operate on oxygen in the pressure range from 50 to 80 pounds per square inch gage (psig). A means shall be provided to maintain correct pressures within the helmet and suit while ascending, descending, or maintaining altitude above that corresponding to 145 plus or minus 5 millimeters of mercury (MM Hg). The regulator shall be designed so that "G" forces, up to 8.67 "G's", will have a minimum effect on the performance of the regulator.

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3.4.1 Plugs.- The inlet and outlets of the regulator shall be closed with caps or plugs to prevent dust and foreign matter from entering the regulator during shipping and storage.

3.5 Construction.- The regulator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Oxygen Inlet Filter.- An oxygen inlet filter equal to and interchangeable with Moraine Products Company, Dayton, Ohio, grade 3 or 4, sintered bronze, shall be provided at the oxygen inlet to prevent foreign particles from entering the regulator.

3.5.2 Relief Valve.- Relief valves shall be incorporated in the body of the regulator to relieve excessive pressure building up at the helmet and suit outlet of the regulator.

3.5.3 Test Pressure Control.- A test pressure control shall provide the suit and helmet with positive pressure at altitudes where automatic pressure is not delivered by the regulator. The control shall provide a pressure of 40 to 60 mm Hg gage pressure for leak testing the oxygen helmet. The suit shall be provided simultaneously with a pressure of five and six times as great as the helmet pressure.

3.5.4 Test Pressure Control Line.- The regulator shall include a test pressure control line so that pressures cited in 3.5.3 may be remotely controlled.

3.5.5 Outlets.- There shall be two separate outlets on the regulator. The helmet outlet shall be legibly marked "Helmet" and the suit outlet "Suit".

3.5.6 Inlet.- The inlet shall be threaded with 1/8 inch female pipe threads.

3.5.7 Orientation.- The regulator shall function as specified regardless of orientation.

3.6 Performance

3.6.1 Outlet Pressures.- The outlet pressures and pressure ratios shall be in accordance with table I with an inlet pressure between 50 and 80 psig.

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TABLE I

Altitude (Feet)	Altitude (MmHg)	Outlet Pressures - Gage: Zero L/M	10 L/M	90 L/M	Ratio of Suit Pressure to Helmet Pressure at 10 L/M	Suit Pressure (In. H_2O Max.)
Ground Level		2.5 to 4.0 In. H_2O	2.0 to 4.0 In. H_2O	-1.0 to 1.0 In. H_2O	- - -	1
10,000	522.6	2.5 to 4.0 In. H_2O	2.0 to 4.0 In. H_2O	-1.0 to 4.0 In. H_2O	- - -	1
35,000	178.7	2.5 to 4.0 In. H_2O	2.0 to 4.0 In. H_2O	0.5 to 4.0 In. H_2O	- - -	1
42,000	127.9	12.1 to 27.1 Mm Hg	12.1 to 22.1 Mm Hg	7.1 to 27.1 Mm Hg	3.0 to 7.0	- - -
46,000	105.6	34.4 to 49.4 Mm Hg	34.4 to 44.4 Mm Hg	29.4 to 49.4 Mm Hg	5.0 to 6.0	- - -
50,000	87.3	52.7 to 67.7 Mm Hg	52.7 to 62.7 Mm Hg	47.7 to 67.7 Mm Hg	5.0 to 6.0	- - -
60,000	54.1	85.9 to 100.9 Mm Hg	85.9 to 95.9 Mm Hg	80.9 to 100.9 Mm Hg	5.0 to 6.0	- - -
70,000	33.6	106.4 to 121.4 Mm Hg	106.4 to 116.4 Mm Hg	101.4 to 121.4 Mm Hg	5.0 to 6.0	- - -

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3.6.2 Venting - Excess Pressure.- Relief valves shall open to vent breathing pressures in the range of 155 to 180 mm Hg and suit pressure in the range of 18 to 24 psig. Prior to venting pressures, they shall not leak in excess of 0.02 liter per minute.

3.6.3 Overall Leakage.- The regulator shall not leak more than 0.5 standard liter per minute when 50 to 80 psig pressure is applied to the regulator inlet, with the outlets plugged and the test control activated.

3.6.4 Overload.- The regulator shall be capable of withstanding a pressure of 180 mm Hg in the helmet outlet, and 950 mm Hg, plus or minus 25 mm Hg, in the suit outlet for 2 minutes each. The maximum pressure relief valve shall be removed or suitably capped for this test. Upon completion of the test, the regulator shall be subjected to the individual tests.

3.6.5 Low Temperature-High Temperature Operation (Cyclic).-- The regulator shall be capable of operation as specified in 4.4.7 while being subjected to a temperature of minus 65 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, and a temperature of plus 160 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit.

3.6.6 Low Temperature Exposure.- The regulator, after exposure to a temperature of minus 85 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, for a period of 48 hours, shall be subjected to and meet the individual tests.

3.6.7 High Inlet Pressure.- The regulator shall meet the requirements of table I after being subjected to an inlet pressure of 120 psig, minus 0, plus 5 psig, for not less than 2 minutes.

3.6.8 Decompression Characteristics.- The regulator shall provide, within 10 percent, the required helmet and suit pressures at the final altitude in not more than 5 seconds when the altitude is changed within 1 second from 15,000 to 50,000 feet and shall provide the pressure cited in table I at 9 seconds. The regulator shall vent any excessive pressure to comply with this requirement.

3.6.9 Endurance and Vibration.- The regulator shall be capable of operating without failure for 180,000 cycles of operation with a peak flow of 30 liters per minute and for 20,000 cycles with a peak flow of 70 liters per minute while being subjected to the vibration requirements of 4.4.11.

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3.7 Threads

3.7.1 Screw Threads.- Unless otherwise specified, the threads for all machine screws 0.06 inch or larger in diameter, shall conform to MIL-S-7742.

3.7.2 Pipe Threads.- Pipe threads shall be in accordance with MIL-P-7105.

3.8 Finish

3.8.1 Aluminum and Aluminum Alloy Finishes.- Aluminum and aluminum alloy parts shall be anodized in accordance with MIL-A-8625 or MIL-C-5541.

3.9 Identification of Product

3.9.1 Nameplate.- A nameplate, permanently and legibly marked in accordance with MIL-STD-130, shall be securely attached to the regulator.

3.9.2 Regulator Marking.- "SUIT" shall be stamped into the body of the regulator near the suit outlet and "HELMET" shall be stamped into the housing near the helmet outlet. The letters shall be approximately 3/16 inch high and shall be filled with white paint.

3.10 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manuracturer's part numbers shall be governed by the drawing number requirements of MIL-D-5028.

3.11 Workmanship.- The regulator assembly shall be manufactured in a thoroughly workmanlike manner. The finished regulators and all integral parts shall be clean and free of oils, greases, and all other materials which might adversely affect the operation of the regulator.

3.12 Date.- The date of manufacture, consisting of month and year shall be legibly stamped with ink by the inspector on the regulator in a place where it will not be defaced during usage. This shall be accomplished with a suitable rubber stamp made in the form of a 1/2-inch square with a horizontal line dividing the square into two segments. The upper segment shall contain the number of the month and the lower segment shall contain the last two numbers of the year. Application of date of manufacture shall certify that all rubber materials used in the regulator do not exceed cure date as specified in 3.3.3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of regulators shall be classified as follows:

a. Acceptance Tests . . . . . .See 4.2
b. Preproduction Testing . . . . . See 4.5

4.2 Acceptance Tests.- Acceptance tests shall consist of:

a. Individual Tests
b. Sampling Plan and Tests

4.2.1 Individual Tests.- Each regulator shall be subjected to the following tests as described under 4.4 Test Methods of this specification:

a. Examination of Product
b. Outlet Pressures
c. Test Pressure Control
d. Venting - Excess Pressure
e. Overall Leakage

4.2.2 Sampling Plan and Tests

4.2.2.1 Lot.- A lot shall consist of 500 regulators or fraction thereof, under essentially the same conditions and submitted for inspection at substantially the same time.

4.2.2.2 Sampling Plan.- Three regulators shall be selected at random from each lot or fraction thereof and subjected to the following tests:

a. Overload
b. Low Temperature - High Temperature Operation (Cyclic)
c. Low Temperature Exposure

4.2.2.2.1 Rejection and Retest.- When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

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4.2.2.2.1.1 Individual Tests May Continue.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3 Test Conditions

4.3.1 Atmospheric Conditions.- Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 degrees Fahrenheit, plus or minus 18 degrees Fahrenheit, and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.3.2 Gas.- The high pressure gas used in testing the regulators shall be either oxygen conforming to BB-O-925, type I, grade A, or water-pumped nitrogen conforming to MIL-N-6011, or water-pumped air equivalent in dryness to MIL-N-6011. If nitrogen or air is used, appropriate density correction factors shall be applied to the flow meter used, to correct not only the effect on the meter itself, but also on the performance of the regulator with the lower density gas.

4.3.3 Adjusted Inlet Pressure.- The inlet oxygen pressure shall be adjusted to between 50 and 80 pounds per square inch gage during all acceptance tests, except where otherwise specified.

4.3.4 Vibration and Noise.- Vibration, flutter, or chatter occuring during breathing or simulated breathing is cause for rejection of the regulator. If this occurs on a sample representing a lot of regulators, the entire lot shall be rejected. If during the testing, the regulator vibrates and the vibration can be stopped by changing or varying the flow and returning to the test point, the regulator is acceptable. If vibration persists after several attempts or if vibration can be obtained by simulating a breathing cycle, the regulator is not acceptable.

4.3.5 Measurement of Suit and Helmet Pressure.- The helmet and suit pressures specified in this specification are those that shall be available to the helmet and suit. Allowances shall be provided to allow for pressure drops in hoses, fittings, and disconnects, between the regulator outlets and the helmet and suit. These allowances shall be added to the pressure specified in this specification to arrive at the values obtainable at the helmet and suit outlets of the regulator.

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4.4 Test Methods

4.4.1 Examination of Product.- Regulators shall be examined to determine conformance to this specification with respect to material, workmanship, dimensions, and marking.

4.4.2 Outlet Pressure.- With the inlet pressure set between 50 and 80 psig, the helmet and suit pressures shall be determined for each altitude and each flow rate specified in table I. The pressures and pressure ratios shall be as specified in table I.

4.4.3 Test Pressure Control.- A pressure between 50 and 80 psig shall be applied to the regulator inlet. With the test pressure control actuated, the helmet outlet pressure shall be 40 to 60 mm Hg gage pressure and the suit outlet pressure shall be between 5 and 6 times the helmet gage pressure at zero flow.

4.4.4 Venting - Excess Pressure.- Relief valves shall open to vent breathing pressures in the range of 155 to 180 mm Hg and suit pressure in the range of 18 to 24 psig. Prior to or after venting pressure, they shall not leak in excess of 0.02 liter per minute.

4.4.5 Overall Leakage.- With the suit and helmet outlets of the regulator plugged, the test control activated, and a pressure between 50 and 80 psig applied to the inlet of the regulator, leakage in excess of 0.5 standard liter per minute shall be cause for rejection.

4.4.6 Overload.- The regulator shall be capable of withstanding a pressure of 180 mm Hg in the helmet outlet, and 950 mm Hg, plus or minus 25 mm Hg, in the suit outlet of the regulator for a period of 2 minutes each. Venting of the helmet pressure relief valve is permissible during the pressure phase of the test. These tests shall not be detrimental to the regulator's operation or components. Upon completion of the test, the regulator shall be subjected to and meet the individual tests.

4.4.7 Low Temperature - High Temperature Operation (Cyclic).- With a pressure between 50 and 80 psig applied to the regulator inlet, the regulator shall be subjected to a temperature of minus 65 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, for not less than 3 hours. Upon completion of the 3-hour period, but while still being maintained at the specified temperature, the regulator shall be subjected to cyclic operation for not less than 2 minutes. The cyclic operation shall consist of approximately 10 cycles per minute at a flow from 0 to 70 liters per minute and back to 0 liters per minute. Upon completion of cyclic operation, the regulator shall be checked to determine conformance to table I. The above test shall be repeated with the regulator stabilized at plus 160 degrees Fahrenheit, plus or minus

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5 degrees Fahrenheit, after exposure to this temperature for not less than 7 hours. An increase of 20 mm Hg in the helmet pressure over that required at altitude above 40,000 feet is permissible for the low temperature test only.

4.4.8 Low Temperature Exposure.- The regulator shall be subjected to a temperature of minus 85 degrees Fahrenheit, plus or minus 5 degrees Fahrenheit, for a period of 48 hours. After the 48-hour period, the regulator shall be allowed to return to room temperature and shall be subjected to and meet the individual tests specified herein.

4.4.9 High Inlet Pressure.- The regulator shall be subjected to an inlet pressure of 120 psig, minus 0, plus 5 psig, for not less than 2 minutes. This test shall not be detrimental to the regulator's operation or components. Upon completion of the test, the regulator shall be subjected to the outlet pressure test.

4.4.10 Decompression Characteristics.- One hundred percent, plus or minus 10 percent of the suit and helmet pressure required at the final altitude shall be attained within 5 seconds when the altitude is changed within 1 second from 15,000 to 50,000 feet, plus or minus 1,000 feet. At 9 seconds, the pressure shall be as cited in table I. The helmet system volume shall be simulated by a 7-liter container at the end of a 3/8-inch hose 26 inches long. The suit system shall be simulated by a 9-liter container at the end of a 3/8-inch hose 26 inches long.

4.4.11 Endurance and Vibration.- With a 70 psig inlet pressure, a peak flow of 30 liters per minute shall be delivered by the regulator for 180,000 cycles of operation. The regulator shall then be operated for 20,000 cycles with a peak flow of 70 liters per minute. During this endurance test, the regulator shall be mounted on a vibration stand for 144 hours operating at a frequency ranging from 500 to 2,500 cycles per minute, with a double amplitude between 0.018 and 0.020 inch. The remainder of the endurance test shall be conducted without vibration. Upon completion of the test, the regulator shall be subjected to the individual tests.

4.5 Preproduction Testing

4.5.1 Preproduction Test Sample Tested by the Contractor.- The contractor shall subject two regulators to the preproduction test specified in 4.5.4.

4.5.2 Preproduction Test Report.- After the contractor completes the preproduction tests, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete

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copies of the report to the procuring activity.

4.5.3 Preproduction Test Sample For the Procuring Activity.- Along with the preproduction test report, the contractor shall submit two samples to the procuring activity.

4.5.4 Preproduction Tests.- Preproduction tests shall consist of all tests described under Test Methods.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-P-7936. Preservation for level A shall be in accordance with method IA, MIL-P-116.

5.2 Marking

5.2.1 Interior and Exterior Containers.- The interior and exterior containers shall be marked in accordance with MIL-STD-129, and as follows:

IF IN STORAGE AFTER * _____ THE REGULATOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE.
*Storage shall be 36 months from date of inspection if a metal container is used, and 18 months if other than metal is used.

5.2.1.1 The following precautionary marking shall be placed on two opposite sides of each interior package:

"FRAGILE - HANDLE WITH CARE"

5.2.1.2 All containers shall be stamped to show the date or regulator manufacture.

6. NOTES

6.1 Intended Use.- The automatic pressure breathing oxygen regulator is intended for use in aircraft at altitudes up to 70,000 feet.

6.2 Ordering Data.- Procurement documents should specify:

a. Title, number, and date of this specification.
b. Preproduction requirements.
c. Level of preparation for delivery.

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